

```
!pip install ultralytics
!pip install deep-sort-realtime
!pip install opencv-python-headless
```

Collecting ultralytics

Downloading ultralytics-8.4.71-py3-none-any.whl.metadata (41 kB)
41.3/41.3 kB 1.3 MB/s eta 0:00:00

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.11.0+cpu)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.26.0+cpu)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)

Collecting nvidia-ml-py>=12.0.0 (from ultralytics)

Downloading nvidia_ml_py-13.610.43-py3-none-any.whl.metadata (9.7 kB)

Collecting ultralytics-thop>=2.0.18 (from ultralytics)

Downloading ultralytics_thop-2.0.20-py3-none-any.whl.metadata (14 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultra
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultra
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultra
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultral
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ul
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultral
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultra
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultra
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ul
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultral
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.8.0->ultra
Downloading ultralytics-8.4.71-py3-none-any.whl (1.3 MB)

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Downloading nvidia_ml_py-13.610.43-py3-none-any.whl (53 kB)

53.2/53.2 kB 3.2 MB/s eta 0:00:00

Downloading ultralytics_thop-2.0.20-py3-none-any.whl (28 kB)

Installing collected packages: nvidia-ml-py, ultralytics-thop, ultralytics

Successfully installed nvidia-ml-py-13.610.43 ultralytics-8.4.71 ultralytics-thop-2.0.20

Collecting deep-sort-realtime

Downloading deep_sort_realtime-1.3.2-py3-none-any.whl.metadata (12 kB)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from deep-sort-realtime) (2.0.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from deep-sort-realtime) (1.16.3)

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (from deep-sort-realtime) (4.13.0)

Downloading deep_sort_realtime-1.3.2-py3-none-any.whl (8.4 MB)

8.4/8.4 MB 22.7 MB/s eta 0:00:00

Installing collected packages: deep-sort-realtime

Successfully installed deep-sort-realtime-1.3.2

Requirement already satisfied: opencv-python-headless in /usr/local/lib/python3.12/dist-packages (4.13.0.92)

```
import cv2
import numpy as np
from ultralytics import YOLO
from deep_sort_realtime.deepsort_tracker import DeepSort
from google.colab.patches import cv2_imshow
```

Creating new Ultralytics Settings v0.0.6 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com>

```
model = YOLO("yolov8n.pt")
```

```
tracker = DeepSort(max_age=30)
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
video_path = list(uploaded.keys())[0]
```

Choose Files No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving Untitled5.ipynb - Colab and 5 more pages - Personal - Microsoft Edge 2026-06-19 09-32-45.mp4 to Untitled5.ipynb - Colab

```
cap = cv2.VideoCapture(video_path)

width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

out = cv2.VideoWriter(
    'output.mp4',
    cv2.VideoWriter_fourcc(*'mp4v'),
    fps,
    (width, height)
)

while cap.isOpened():
    ret, frame = cap.read()

    if not ret:
        break

    results = model(frame)

    detections = []

    for r in results:
        boxes = r.boxes

        for box in boxes:
            x1, y1, x2, y2 = map(int, box.xyxy[0])

            conf = float(box.conf[0])
            cls = int(box.cls[0])

            label = model.names[cls]

            detections.append(
                ([x1, y1, x2-x1, y2-y1], conf, label)
            )

    tracks = tracker.update_tracks(
        detections,
        frame=frame
    )

    for track in tracks:
        if not track.is_confirmed():
            continue

        track_id = track.track_id

        l, t, r, b = map(int, track.to_ltrb())

        cv2.rectangle(
            frame,
            (l, t),
            (r, b),
            (0, 255, 0),
            2
        )

        cv2.putText(
            frame,
            f"ID: {track_id}",
            (l, t-10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.7,
            (0, 255, 0),
            2
        )

    out.write(frame)

cap.release()
out.release()

print("Processing Complete!")
```

0: 352x640 1 laptop, 204.8ms
Speed: 5.6ms preprocess, 204.8ms inference, 33.1ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 210.8ms
Speed: 8.0ms preprocess, 210.8ms inference, 1.7ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 206.2ms
Speed: 6.6ms preprocess, 206.2ms inference, 1.4ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 208.2ms
Speed: 6.2ms preprocess, 208.2ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 206.0ms
Speed: 7.3ms preprocess, 206.0ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 196.7ms
Speed: 6.2ms preprocess, 196.7ms inference, 1.6ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 206.8ms
Speed: 5.4ms preprocess, 206.8ms inference, 1.6ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 203.4ms
Speed: 5.0ms preprocess, 203.4ms inference, 1.5ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 204.2ms
Speed: 7.9ms preprocess, 204.2ms inference, 1.6ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 165.5ms
Speed: 8.7ms preprocess, 165.5ms inference, 1.2ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 135.2ms
Speed: 5.1ms preprocess, 135.2ms inference, 1.3ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 129.7ms
Speed: 3.5ms preprocess, 129.7ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 130.4ms
Speed: 3.2ms preprocess, 130.4ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 129.2ms
Speed: 3.2ms preprocess, 129.2ms inference, 1.2ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 131.8ms
Speed: 5.1ms preprocess, 131.8ms inference, 1.0ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 139.0ms
Speed: 3.4ms preprocess, 139.0ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 130.7ms
Speed: 3.6ms preprocess, 130.7ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 129.5ms
Speed: 3.7ms preprocess, 129.5ms inference, 1.0ms postprocess per image at shape (1, 3, 352, 640)

0: 352x640 1 laptop, 128.6ms
Speed: 3.0ms preprocess, 128.6ms inference, 1.1ms postprocess per image at shape (1, 3, 352, 640)

```
from google.colab import files  
  
files.download("output.mp4")
```

```
cap = cv2.VideoCapture(0)
```

```
cap = cv2.VideoCapture(video_path)
```

